Introduction to Artificial Intelligence

Prof. Dr. Hamed Shariat Yazdi, Darmstadt University of Applied Sciences

Assignment: Knowledge Representation

In this lab session, you will practice SPARQL with the knowledge graph TMDB.ttl (download from Moodle course). You will be working in your teams of two students each.

1. Read Sections 3.3 and 3.4 of the book "Applied Artificial Intelligence – An Engineering Approach".
2. Answer the quick check questions
3. Download TMDB.ttl from the Moodle course
4. Install Fuseki from https://jena.apache.org/documentation/fuseki2
5. Start the Fuseki server;
   Open Fuseki in http://localhost:3030
   Add a new dataset "TMDB"
   Upload TMDB.ttl
6. Implement SPARQL queries and test them in Fuseki. Document the SPARQL queries as well as screenshots of the results in a file. Store the file in PDF format in the Git repository. The SPARQL queries shall answer the following questions:
   a. Which genres exist?
   b. Which departments exist for crews?
   c. How many companies are listed?
   d. Show all collections
   e. List German movies
   f. Show movies in which Julius Caesar is depicted
   g. Which movie actors also directed a movie?
   h. All roles of your favourite actor
   i. All details about your favourite movie
   j. More interesting questions of your choice, e.g., most popular movies / actors / genres, …
7. OPTIONAL: try out some SPARQL queries in Python using rdflib.

8. Commit your file with SPARQL queries and (optionally) your Python code to the Git repository